SKILL BUILDERS · UNIT 4 · SCORE 3

Angle Relationships: Mini-Lessons

NC.8.G.5

Name _____ Date _____ Period _____

4.1 Angle pairs NC.8.G.5

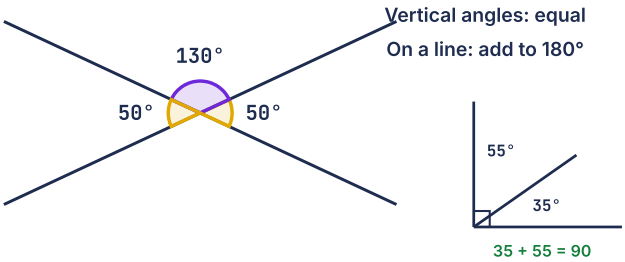
Angles come in pairs that follow rules. **Complementary** angles add to **90°**. **Supplementary** angles add to **180°**. **Vertical** angles (across from each other where two lines cross) are **equal**.

Complementary
two angles that add to 90° (a corner)

Supplementary
two angles that add to 180° (a straight line)

Vertical angles
across from each other at a crossing: equal

Adjacent
next to each other, sharing a side



Watch out: Mixing up 90 and 180?
Complementary = **C**orner = 90°.
Supplementary = **S**traight = 180°.

Walk through: `Two angles are supplementary. One is 110°. Find the other.`

1 Supplementary angles add to...

_____

2 How do we find the missing angle?

_____

3 The other angle is...

_____

Check your understanding

1 Score 3 NC.8.G.5

Find the **complement** of 62°.

Answer: _____ °

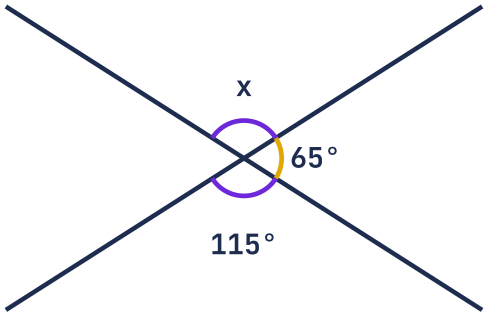
2 Score 3 NC.8.G.5

Find the **supplement** of 145°.

Answer: _____ °

3 Score 3 NC.8.G.5

Two lines cross. Find **x**.



Answer: _____ °

4.2 Parallel lines cut by a transversal NC.8.G.5

When a transversal cuts two **parallel** lines, every angle is either **equal** to a given angle or **adds with it to 180°**. Equal pairs: **corresponding, alternate interior, alternate exterior, vertical**. Same-side interior angles add to **180°**.

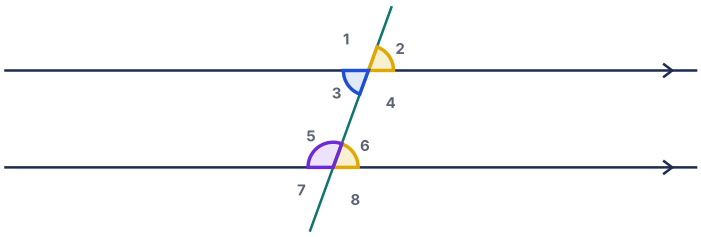
Transversal
a line that crosses two other lines

Corresponding
same corner at each crossing: equal

Alternate interior
between the lines, opposite sides: equal

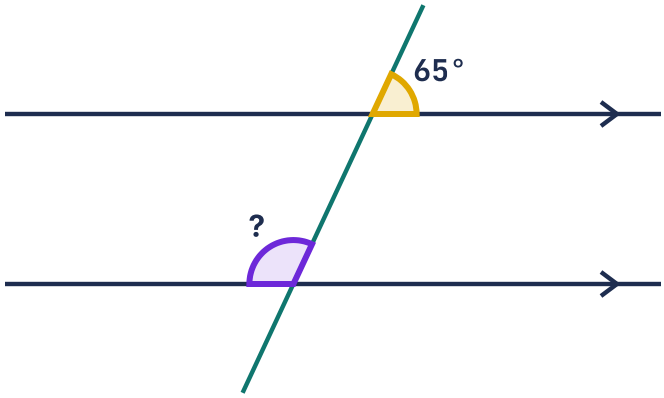
Same-side interior
between the lines, same side: add to 180°

Watch out: Look at the picture: a small-looking angle can only equal another small-looking angle.



small = 70°, big = 110°

Walk through: Lines are parallel. ∠2 = 65°. Find ∠5.



1 Is ∠5 small-looking (acute) or big-looking (obtuse)?

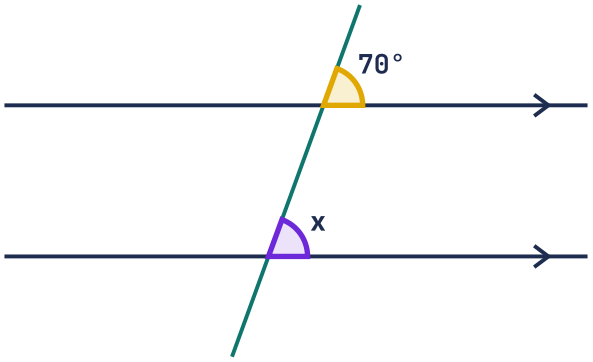
2 So ∠5 and ∠2 are...

3 Find ∠5.

Check your understanding

1 Score 3 NC.8.G.5

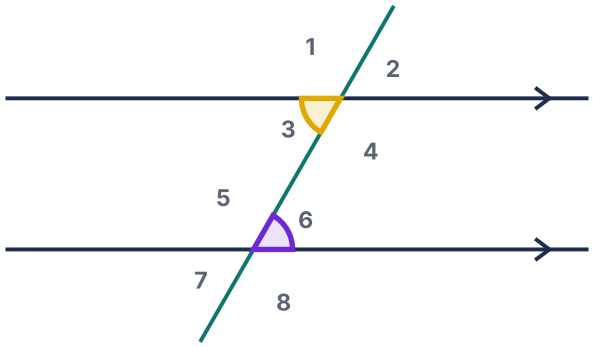
The lines are parallel. Find **x**.



Answer: _____ °

2 Score 3 NC.8.G.5

What kind of pair are ∠3 and ∠6?



(A) Corresponding angles

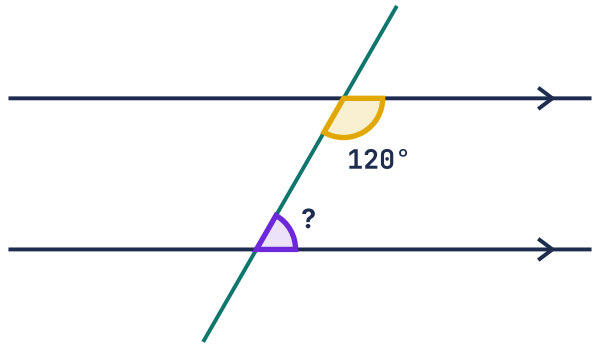
(B) Alternate interior angles

(C) Vertical angles

(D) Same-side interior angles

3 Score 3 NC.8.G.5

The lines are parallel. ∠4 = 120°. Find ∠6 (a same-side interior angle).



Answer: _____ °

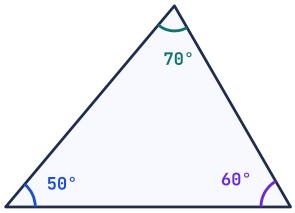
4.3 Angles in a triangle NC.8.G.5

The three inside angles of **any** triangle add to **180°**. To find a missing angle: add the two you know, then subtract from 180.

Interior angle
an angle inside the triangle

Right triangle
a triangle with one 90° angle (a square corner)

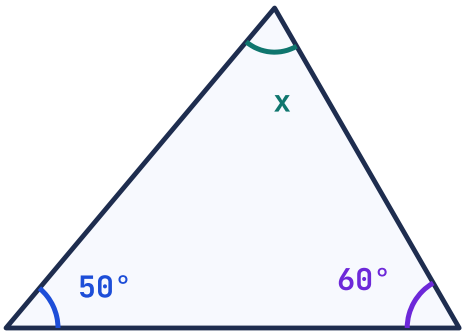
Watch out: Use 180°, not 360°. 360° is a full circle (or a four-sided shape).



50 + 70 + 60 = 180



Walk through: `A triangle has angles 50° and 60°. Find the third angle.`



1 The three angles of a triangle add to...

_____

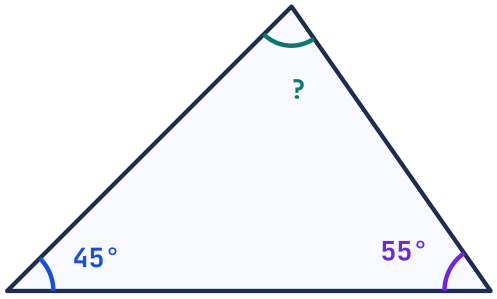
2 Add the angles you know.

3 Find x.

Check your understanding

1 Score 3 NC.8.G.5

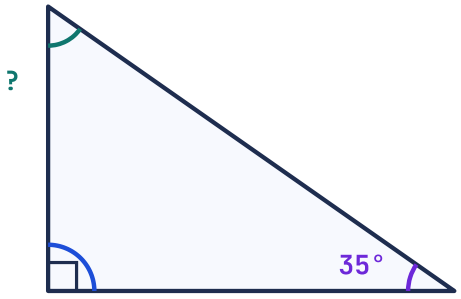
A triangle has angles 45° and 55°. Find the third angle.



Answer: _____ °

2 Score 3 NC.8.G.5

A right triangle has one angle of 35°. Find the other small angle.



Answer: _____ °

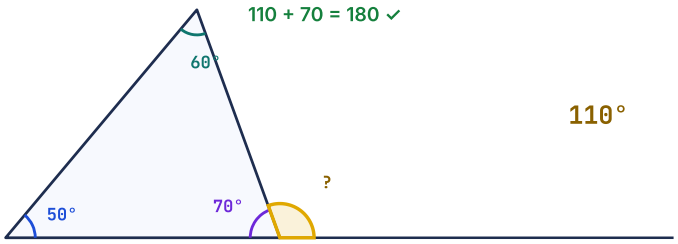
4.4 The exterior angle of a triangle NC.8.G.5

An **exterior angle** is outside the triangle, made by stretching one side. It **equals the two far inside angles added together**.

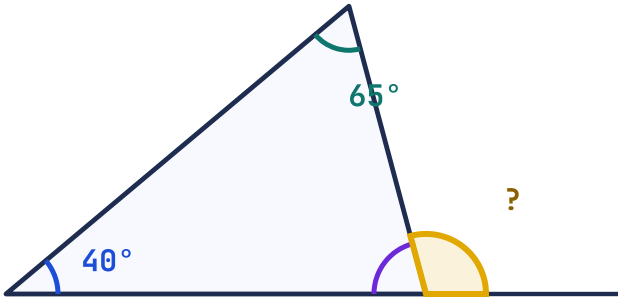
Exterior angle
the angle outside the triangle, next to one inside angle

Remote (far) interior angles
the two inside angles that do not touch the exterior angle

Watch out: Use the two **far** angles, not the angle right next to the exterior angle.



Walk through: `The far inside angles are 40° and 65°. Find the exterior angle.`



1 Which angles do we use?

_____

2 Add them.

_____

Check your understanding

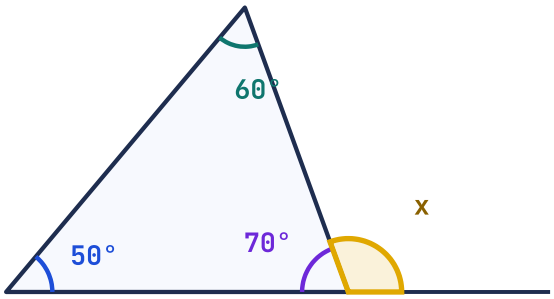
1 Score 3 NC.8.G.5

The far inside angles are 40° and 65°. Find the exterior angle.

Answer: _____ °

2 Score 3 NC.8.G.5

Find the exterior angle **x**.



Answer: _____ °